Research reader and verification contract

Questions, definitions, evidence and calibrated trust

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Repository links point to current main for navigation, not historical provenance.

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Project-defined process guidance, not a PID result or proof. This contract extends the existing mathematical workflow without replacing its claim packets, negative records or historical publications. Any portable document format (PDF) version follows the same evidence boundaries.

1 Question and strategy

The research question is: which statements about a named shared-exclusions functional can be derived correctly, computed reliably, and used for a justified decision? These are separate questions. Name the defining publication before using its PID (partial information decomposition). Categorical Makkeh–Gutknecht–Wibral shared exclusions and continuous Ehrlich shared exclusions remain distinct. See the existing source-pinned vocabulary.

Every study opens with five short answers:

1. What exact question is being answered, and why does its answer matter?
2. Which law, functional, inputs, outputs and assumptions define the question?
3. Which strategy was actually used, whether artificial intelligence (AI) assisted or not? State AI's actual role or its absence, and explain the choice against the nearest alternatives: direct algebra, finite enumeration, ordinary tests, a solver, a simpler non-PID analysis or no action.
4. Which valuable result survives checking, which evidence supports it, and what application decision follows—or does not follow?
5. What failed, what remains open, and what evidence would justify reconsideration?

Record the actual strategy, not an ideal strategy retrospectively assigned to the run. AI may generate candidates, reconstruct arguments, find counterexamples, inspect sources or review exposition. For decisive claims, an independent-first route freezes its initial derivation before seeing another route's candidate. After exposure, it is an adversarial review, not an unexposed reconstruction. Keep shared source, model, parser, runtime and custody dependencies visible. State separately what is useful now and what is only a conjectural future use. Name the future test needed; do not count it as present application evidence.

Use the existing claim packet and route memo to map an external method's premises to the named pid-rs need, identify adaptations and broken premises, compare the simplest applicable alternative, and specify a falsifiable acceptance test. Preserve rejected transfers. A source's benchmark, reviewer score or different-domain success gives no PID correctness, calibration, novelty or application credit.

2 Cited glossary

Use this compact process glossary alongside each study's own complete notation. Its definitions follow the linked existing repository contracts; no new method is introduced.

Term	Meaning and defining reference
Estimand	The exact quantity a procedure aims to estimate; method boundaries
Functional	A map from a declared probability law to a quantity; reading conventions
Empirical law	Relative frequencies of complete observed categories; not automatically the population law; finite-law contract
Independent-first route	An initial reconstruction frozen before exposure to another candidate; not statistical or institutional independence; route dependencies
Shared cut	A dependency whose failure can defeat several nominally different routes; evidence architecture
Semantic correspondence	The justified link between intended meaning and a formula, formal statement or executable; formal boundaries
Calibration	Agreement between stated statistical guarantees and performance under the declared repeated-sampling model; sampling boundaries
Nonclaim	A conclusion the evidence must not be used to support; evidence card
Negative control	An intentionally invalid test input that the declared acceptance rule must reject; attack matrix
Counterexample	An admissible example that satisfies a claim's premises but violates its conclusion; not merely an invalid input; falsification workflow
Proof kernel	The component that checks encoded proofs under the permitted axioms; not an estimator's neighbourhood rule; tool boundaries
Custody	Recorded preservation and recovery of identified evidence; not its truth or authorship; durability rules

3 Explain a formula before asking the reader to use it

Before each central formula, define its symbols locally. Give each variable's meaning, domain, index set and units. Identify source and target roles, the law used by each expectation, support and sampling conditions, and every required positive denominator. Define abbreviations at first use. A glossary and a bibliography do not replace local definitions.

For example, let n be a positive integer counting complete observed rows. Let z be one joint source-and-target category in a declared finite alphabet, and let c_z be its nonnegative integer row count. The empirical probability of that category is $\hat{p}_n(z) = c_z/n$. Its unit is dimensionless, and the counts must sum to n. If $n = 4$ and $c_z = 1$, the probability is $1/4$. This arithmetic evaluates an empirical law; it does not show how well that law estimates a population law. A new equation needs its own local meanings when a symbol's role changes.

Show the intermediate steps that carry the conclusion: substitutions, normalizations, signs, index changes, inequalities, cancellations and error aggregation. Use a small exact example that a reader can recompute. Separate an observed physical input from a synthetic mathematical example. An angle in radians does not establish a particular angular density model.

4 Evidence and trust

For each important conclusion, record the narrow supported statement and its strongest nonclaim. Keep these evidence classes separate:

Evidence	What it supports	What it does not supply
Published definition	The named source's mathematical object	Correct local transcription or implementation
Analytic derivation	The displayed implication under its stated premises	Proof-kernel acceptance, runtime behavior or calibration
Formal replay	The recorded encoded statement under its imports and axioms	Intended scientific meaning or unrecorded descendants
Exact finite check	The stated finite predicate on the checked domain	A universal analytic or sampling theorem
Executable comparison	The bound implementations agree on the workload	Correctness outside the workload
Statistical evaluation	The recorded study under its design and uncertainty assumptions	Deployment, causal value or an untested population
Actual PDF review	The viewed pages at the recorded sizes	Mathematical truth or unviewed page approval
Human verification	Only the named person's recorded checks	Review that was never performed

Do not label a model's manual algebra as human verification. When no human verification receipt exists, state that it is absent. Trust follows the load-bearing evidence and its shared dependencies, not confidence, votes, reviewer counts, polished language or a green build. The current stock-Lean promotion hold remains applicable; a new process document cannot remove it. Historical receipts are not thereby erased or disproved.

5 Failure and originality records

Preserve positive and negative results with exact assumptions, decisive examples, evidence, rejection reasons and conditions for reconsideration. Correct an active claim without erasing its false

predecessor. Link a dated erratum or successor from the current result map. Keep immutable receipts immutable.

Before claiming an advance, search for equivalent statements, reparameterizations, special cases and compositions of known results. Record the search date, sources, queries or theorem locators, nearest prior result and the exact additional claim. “No equivalent found within this search” is not proof of global novelty.

A small searched publication corpus does not establish what a model encountered during training. Training exposure is unknown unless independently evidenced. A locally withheld example is not necessarily unseen in pretraining. Model agreement cannot answer whether an intermediate result was recalled or derived. Reconstruction, counterexamples and correspondence checks address correctness; dated equivalent-theorem searches address the narrower originality question.

6 Extension precedence for retained wording

This extension supersedes only the following two retained sentences in the mathematical workflow. All other claim packets, mathematical meanings, evidence requirements and historical receipts retain their existing authority. No new formal, numerical or application credit follows.

- Under “Codex goal, plan, and tool lifecycle,” the retained sentence is: “A paused, hard, slow, or partially complete goal stays active.” Correct its paused-goal case: an explicit user request to pause requires `update_goal(status="paused")` when that tool is available, reporting its returned state and stopping goal work. Use this transition only at the user’s explicit request. Difficulty, slowness or partial progress alone is not a pause request. A resumed blocked goal starts a fresh blocked audit. Use plan or goal tools only when actually available; never invent a successful call.
- Under “Optional explicit mapping-check rule,” the retained sentence is: “Every categorical claim must answer five questions in its claim packet:” Here “categorical claim” means a **category-theoretic claim**. It does not mean a claim about finite category-valued data. The five questions and their existing mathematical obligations are unchanged.

The older publication remains a dated preserved artifact. Updating its complete Markdown/TeX/PDF projection and visual receipt is a separate publication task; this contract does not claim that task is done.

7 Validation and implementation

A structural checker can verify row counts, declared line/page coverage, links, byte hashes and declared source/PDF projection bindings. These checks establish record consistency; they cannot prove that a reported reading actually happened, that every symbol is meaningful, that the projection preserves meaning or that every premise is scientifically true. Semantic review, mathematical checks, qualified formal replay, statistical evaluation and actual page inspection retain separate receipts. Test missing premises, undefined domains, stale bindings, erased negatives and unjustified review labels, with harmless valid changes as positive controls. A finite mutation suite remains finite.

This process contract changes no Rust API, PID estimator, proof kernel, toolchain, tolerance or scientific acceptance predicate. No timing or application benefit is reported. New scientific studies still require the existing “Rust implementation and computational cost” section.

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